

Borders and Approvals

A product can be legal in one country, restricted in another and unknown in a third, and the map of who approved what is itself a piece of scientific evidence

There is a sentence in the press materials of nearly every company that has tried to sell a bovine hemoglobin oxygen carrier: it is approved in such-and-such a country. The sentence is true. It is also one of the most misunderstood facts in the field, because an approval is not a universal certificate of quality. It is a decision made by one regulator, under one set of rules, on one body of evidence, for one specified use, at one moment in time.

This essay is about that map. It describes how approval, restriction and exceptional access differ, what the pattern for BHOC products actually looks like, and how a patient community should read a claim of approval elsewhere. I will try to be careful about what I can and cannot confirm, because the details vary with the source and this is a subject on which loose language does real harm.

What an approval is

An approval is a regulator's judgment that a product's benefits outweigh its risks for a defined population under defined conditions. Each part of that sentence hides a variable.

The judgment depends on the evidence submitted. A regulator in a small market may accept data from trials done elsewhere, may require local studies, or may weigh a company's dossier differently from a regulator in a large market with a long history of assessing similar products.

The benefit-risk balance depends on what alternatives exist. A product that would be marginal where safe blood is plentiful may be valuable where it is scarce. Regulators in different countries face different alternatives, and they legitimately reach different conclusions.

The defined population and conditions matter enormously. An approval to treat anemia in adult surgical patients does not authorize use in trauma, in children, or in pregnancy. The label is a fence.

And an approval is a snapshot. It can be revised, restricted or withdrawn. It says nothing about how the product has fared since.

When a reader sees the phrase approved in a country, each of those questions is still open.

The record for the bovine products

Here is what I can state with reasonable confidence, drawing on a company's own description and on published histories, and flagging where these sources differ.

The veterinary product, Oxyglobin, was approved in the United States and Europe in the late 1990s, and its current owner describes it as approved for over twenty years, with more than 240,000 units sold and off-label use in about thirty species. That is a company statement, not an independent audit, but it is consistent with the general history.

The human product, Hemopure, was approved in South Africa in 2001 for the treatment of acute anemia in adult surgical patients. That approval is reported consistently across sources. It was also approved in the Russian Federation. The company's website gives the year as 2010 and describes the indication as acute, all-cause anemia. A published historical review states the same approval without giving a year, and one encyclopedic source reports the approval in a slightly different way. I mention the discrepancy because it illustrates a point: even the basic facts of approval are recounted with variations, and a careful writer should cite the regulator's own record, which I have not seen.

In the United States, Hemopure has never been approved for human use. The Food and Drug Administration placed a hold on a proposed trauma trial in 2003 and did not approve an application for orthopedic surgery that year. Since then the product has been available under the agency's expanded access mechanism, used to treat patients with life-threatening anemia who cannot receive blood. A registered expanded access protocol exists for that purpose.

I am not aware of any approval of a bovine hemoglobin carrier for human use in the European Union, though I would not be surprised to learn of specific national authorizations or named-patient arrangements, and I have not searched for them exhaustively.

Two features of this map stand out. It is small: two countries, one indication. And it was not built through the largest, most demanding regulators. That does not make the approvals meaningless. It means they carry a particular kind of evidence.

Why small-market approvals happen

There are several reasons a company might pursue approval in a smaller market first, and none of them is sinister.

The regulatory path can be shorter and cheaper. A company running short of money has a strong incentive to reach any market where it can start to earn revenue and gather data.

The need can be greater. In a country with a limited blood supply and a high burden of trauma or of diseases that cause anemia, the argument for an alternative is stronger, and a regulator can reasonably weigh it that way. South Africa, with both a high rate of trauma and constraints on blood availability, is a plausible example, although I would want to see the regulator's own reasoning before saying so.

Some regulators have procedures for products that meet an unmet need, allowing conditional or provisional authorization on less complete evidence than the largest agencies require, with post-marketing obligations to gather more.

A first approval can provide a base for further trials. The product can be used in a real health system, adverse events can be recorded, and a supply chain can be tested. In principle, that is valuable. In practice, the value depends on whether the data are actually collected and published.

What an approval elsewhere does and does not tell a regulator here

A frequent question from patient groups is why the United States does not simply accept an approval from another country. The answer is that a regulator's obligations are to its own population and its own standards.

An approval abroad is a piece of evidence. It shows that another agency reviewed the product and found the benefits to outweigh the risks in its context. If the reviewing agency is rigorous, and the data are complete, and the trial population resembles the domestic one, the approval carries weight. American law permits foreign clinical data to support an application under certain conditions, principally that the data come from well-conducted studies and apply to the American population.

But the approval is not a substitute for the application. The regulator here will ask about the same issues it would ask about any product: the quality of the manufacturing, the completeness of the safety database, the adequacy of the trials, and above all the comparison with the standard of care. If the standard of care in the reviewing country differs from the standard here, the comparison may not transfer.

There is also history. The class of products has a record that includes trials with excess deaths and a pooled analysis that raised alarms. A regulator looking at a new application inherits that memory and will look harder at the data than for a product from a class with a cleaner history. This is prudent, not vindictive.

The exceptional routes

Between approval and prohibition lies a range of exceptional access mechanisms, and understanding them is essential for a patient community, because most people who receive an unapproved oxygen carrier do so through one of them.

In the United States the agency's expanded access framework, sometimes called compassionate use, permits a patient with a serious or life-threatening condition, and no satisfactory alternative, to receive an investigational product outside a clinical trial. There are several forms. An individual patient request is made by a treating physician, who obtains the manufacturer's agreement to supply the product and submits a request to the agency. In a true emergency the agency can authorize use by telephone and the paperwork follows. Larger forms cover intermediate-size groups of patients and broader treatment protocols.

Each form requires the agreement of several parties: the manufacturer must be willing to supply, the physician must be willing to take on the responsibility of oversight, the hospital's ethics committee generally must review, and the patient or a surrogate must consent. Any of them can say no, and the product is not owed to anyone.

Other countries have their own arrangements, with names like named-patient supply, special access and authorization for unlicensed medicines. Their rules differ in what they require and how fast they move. In some, a physician can import a product for a single patient with relatively light paperwork. In others, the process is closer to a formal application.

From a patient's point of view, the key features of these routes are speed, cost and certainty. They can be fast, but only if everyone involved knows what to do. They can be expensive, because the manufacturer may charge and the hospital may bear administrative costs. And they are uncertain, because the answer is discretionary.

The evidence problem in exceptional use

Here is the difficulty that connects the map back to the science. The more a product is used through exceptional routes, the less it is studied.

Expanded access is not designed to generate evidence. Patients are not randomized, outcomes are not standardized, follow-up is uneven, and reporting is voluntary. A collection of case reports, however moving, cannot show that a product works. Patients who receive the product are, by definition, in desperate circumstances, and some will die whatever is given. Others will survive for reasons that have nothing to do with the treatment.

A registry that recorded each case in a common format would make the record more useful. It would specify what was collected, in whom, and how outcomes were defined, and it would be reported in aggregate. The FDA has encouraged data collection from expanded access, and companies sometimes publish series of cases, but the result is a scatter rather than a dataset.

I raise this as a reason for patient groups to think of exceptional access as a responsibility as well as a right. A group that helps a patient obtain a product under expanded access might also help ensure that the case is recorded, with consent, in a way that can be pooled with others.

The trial that crosses a border

One event in the recent record shows how the map can be used constructively. A press release announced that a trauma trial in the prehospital setting would be sponsored by the United States Department of Defense and coordinated by Stellenbosch University in South Africa, evaluating a bovine hemoglobin carrier. The arrangement is instructive on its face: an American funder interested in military applications, a South African academic center with experience in trauma care and a product already approved there, and a trial design that could produce data relevant to both.

I have not found a completed report of that trial in the sources I could reach, and I will not describe results I cannot cite. What I can say is that the structure of it is sensible. An approved product in one jurisdiction can be tested under the local approval, in a setting with a genuine need, generating data that might eventually support an application elsewhere. It is a model that other programs could follow, and one a patient organization could encourage.

A note on language

Words on this map are slippery, and patient communications should be exact.

Approved should be followed by a country, an indication, and a year, and by the source of the statement. Available under expanded access should not be shortened to available. Used in South Africa should not be turned into proven safe. Studied in a Department of Defense trial should not be translated into endorsed by the military.

When a company says a product has been used in many patients, ask how many, in what setting, under what protocol and with what follow-up. Numbers without denominators and without outcomes are not evidence.

And when a sentence begins with the phrase the FDA has not approved, it is worth completing it with what the FDA has actually done: held a trial, declined an application, or permitted use under expanded access. Each of those is a different fact.

What a good map would show

If I were drawing the ideal map for the community, it would have six columns for each product and each country: the regulator's name, the year, the indication, the legal status, the type of evidence relied on, and a link to the regulator's own document. It would be maintained by someone with no stake in the outcome. It would state what is unknown.

Such a map does not exist in one place. Building it would be a service to patients, clinicians and journalists alike, and it is well within the capacity of an advocacy organization with the patience to read regulatory records. It would also protect the community against a common failure of the field, which is repeating a favorable sentence until it becomes a belief.

What is inside a dossier

It helps to picture what a regulator actually reads, because the word approval compresses a mountain of paper into one syllable. A submission for a biologic product typically has four large parts.

The first covers quality and manufacturing. It describes the source material, the herd or the donor pool, the steps of the process, the tests done on every batch, the limits set for impurities, and the evidence that the process is consistent from one batch to the next. For a bovine product this is where the animal-health documentation lives, and where the viral and prion clearance studies described in another post are submitted. Many products that are scientifically interesting founder here, because a process that works in a laboratory must be shown to work the same way every time at scale.

The second part is nonclinical: studies in animals and in the laboratory that show what the product does, how long it lasts, which organs it reaches, and what happens at doses well above the intended one. Regulators look especially at the kidney, the heart and the vessels for a hemoglobin product, given the history.

The third is the clinical section, the trials in people. For a product aimed at emergencies it is rarely as tidy as for a drug taken daily for a chronic condition. Trials may be small, may enroll a mixture of patients, and may use comparators that reflect local practice.

The fourth is the proposed labeling and the plan for monitoring after approval: what the leaflet will say, who may use the product, and what data the company promises to collect once it is on the market.

Each part can pass or fail on its own. A company can have persuasive clinical data and be refused because its manufacturing was inconsistent. It can have clean manufacturing and be refused because the trials could not distinguish benefit from chance. When a regulator says no,

the reason can be found in the response letter, and a patient advocate who reads it learns more about the state of the product than from any press release.

A checklist for reading a claim of approval

Suppose you receive a message, a brochure or a news story that says a product is approved somewhere. Here are the questions I would ask, in order.

Who approved it? The name of the regulator, not just the country. Some agencies issue approvals that other authorities regard as thorough. Others act more quickly on less.

For what? The indication as written on the label, in the label's own words. Anemia in surgery, anemia from any cause and trauma are different claims.

For whom? Adults or children, hospital or field, with or without blood available.

When? A decade-old approval may predate newer information, and an approval can be quietly allowed to lapse when a company stops supplying a market.

On what evidence? Randomized trials, uncontrolled series, or a decision based on prior approval elsewhere.

With what conditions? Some approvals come with obligations to run further studies. Did the studies happen?

Is it actually available? A product may be approved and yet not be manufactured, not be stocked, or not be affordable. Approved on paper and available in a hospital are two separate states.

What has happened since? Have there been safety communications, restrictions or withdrawals?

None of these questions requires expertise. Each one converts a comforting phrase into a set of specifics, and the specifics either hold up or they do not.

Why fragmentation costs patients

I want to end the substantive part of this essay on a practical worry. A fragmented map has a cost that falls on patients and that no single regulator is responsible for.

When each country decides independently, developers must repeat work. Trials are designed to satisfy different agencies, files are prepared in different formats, and small companies exhaust their resources on parallel submissions. In a field already short of capital, duplication is lethal.

Regulators in different countries do have mechanisms to cooperate, including shared inspections, mutual reliance on each other's reviews in some circumstances, and international guidelines for the content of dossiers. These tend to work best among large, well-resourced agencies, and less well for the small markets where a niche product might first be approved.

A patient organization cannot rewrite these systems. It can help in two ways. It can push for common data standards, so that the outcome of an emergency use in one country can be read in

another. And it can support trials whose designs are acceptable to more than one regulator, so that a single study can serve several applications. Neither of these is glamorous. Both would reduce the number of patients who wait while paperwork repeats itself.

Closing

Approval is a door, not a verdict. A product that has passed through one door has not thereby passed through all of them, and a product that has not passed through the largest one has not thereby failed. The fair way to read the map is as a set of partial answers to the same underlying question, asked by different people with different stakes.

Further reading

The company's description of its approvals and expanded access program appears on the website of HbO2 Therapeutics. A narrative review of the history of hemoglobin-based oxygen carriers, published in *Annals of Blood*, states the approval of Hemopure in South Africa in 2001 and in Russia and describes its use in the United States under expanded access. Details of the Department of Defense sponsored prehospital trial were announced in a press release from the sponsor and coordinating center. Readers should consult the regulators' own databases for the current status of any product, since approvals change and secondary sources vary.